THE ORDINARY TRACE IS #G TIMES THE G-TRACE ON A FREE RANK-ONE MODULE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file trace.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- Both traces are additive on direct summands of finite projective modules. invoked at proof line 1669. Justifies the reduction to the free rank-one case.

Lemma 0.2. *Let $f : P \to P$ be an endomorphism of the finite projective $\Lambda[G]$-module P. Then*

$$Tr_\Lambda(f; P) = \#G \cdot Tr^G_\Lambda(f; P).$$

Proof. By additivity, reduce to the case $P = \Lambda[G]$. In that case, f is given by right multiplication by some element $\sum \lambda_g \cdot g$ of $\Lambda[G]$. In the basis $(g)_{g\in G}$, the matrix of f has coefficient $\lambda_{g_2^{-1}g_1}$ in the (g_1, g_2) position. In particular, all diagonal coefficients are λ_e, and there are $\#G$ such coefficients. □

Date: stacks-project:trace.tex:local-snapshot-20261012; GNU FDL 1.2 or later.